

Non-prescription pain relievers

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This education module is independently researched and compiled by PSA-commissioned authors and peer reviewed.

Pain relief is one of the most common reasons people visit a pharmacy. Pain can have a significant impact on a person's quality of life including their physical and mental health, and ability to work, sleep, exercise and perform everyday tasks. Pharmacists can play an important role in assisting people to select an appropriate medicine to treat their pain. Pharmacists can also provide advice on the safe and correct use of pain relieving medicines and pain self management strategies; and recognise when referral to a medical practitioner or other healthcare professional is appropriate.



Severe acute pain, nerve injury and psychosocial factors can predispose to the development of chronic pain.

Learning objectives

After reading this article, pharmacists should be able to:

- Describe the differences between acute and chronic pain.
- Discuss potential side effects and important safety precautions for non-prescription pain relievers.
- Advise on the safe and correct use of non-prescription pain relievers.

Competencies addressed (2010): 1.3, 4.2, 6.1, 6.2, 6.3, 7.2

What is pain?

Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage.

Acute pain

Acute pain refers to pain that occurs suddenly and has an identifiable cause associated with injury or tissue damage. It usually has a short duration and is self-limiting. The pain subsides as the injured tissue heals.¹ Common causes of acute pain include:²

- muscular pain from sprains, strains or other sports injuries
- back pain
- headache and migraine
- dysmenorrhoea (period pain)
- toothache.

Acute pain can be recurrent or cyclical and have a similar presentation to chronic pain. Conditions that often cause recurrent episodes of acute pain include headaches and dysmenorrhoea.³

Chronic pain

Chronic pain is a more complex process and is considered a disease in its own right. Pain is classified as chronic (or persistent) when it persists for more than three months and continues after normal healing.¹

Chronic pain often begins as acute pain. Types of acute pain that may progress to chronic pain include post-operative and post-traumatic pain, acute back pain and shingles pain. As acute pain progresses to chronic pain, nerves in the central nervous system become sensitised and respond in an excessive way to potentially harmful (noxious) stimuli and non-noxious stimuli such as touch and light pressure.³ Severe acute pain, nerve injury and psychosocial factors can predispose to the development of chronic pain.^{2,3} Failure to treat acute pain promptly and appropriately at the time of injury can contribute to the development of chronic pain.

Neuropathic pain is a type of pain associated with injury or dysfunction in the nervous system. It is often described as sharp, stinging or stabbing pain, burning

or tingling sensations, or sensitivity to cold or touch.⁴ It is a common cause of chronic pain symptoms, but can present acutely following trauma or surgery. Around 3% of people with acute pain experience neuropathic pain.² Common causes of neuropathic pain include peripheral nerve damage associated with uncontrolled diabetes, and postherpetic neuralgia due to inflammation of nerve tissues caused by herpes zoster (shingles).⁴

Assessment of pain

Pain assessment tools can be used to understand the type and severity of pain a person is experiencing. For pharmacists, these can help to determine the likely cause of the pain, and whether referral is necessary.

A range of pain assessment tools are available. Multidimensional tools assess several aspects of pain. An example is the "L-I-N-D-O-C-A-R-F-F" mnemonic tool which can be used in a pharmacy setting to obtain a more complete picture of the pain a person is experiencing (see Practice Point 1). Single dimensional assessment tools include numeric, verbal or visual scales that rate only one aspect of pain.¹ With numerical scales, people are asked to use numbers from 0 to 10 to rate the intensity of their pain (0 being no pain and 10 being unbearable pain). Verbal scales contain commonly used words such as 'mild,' 'moderate' and 'severe' to help people describe the severity of their pain (see Table 1. in Counter Connection).

Visual scales use aids such as pictures of facial expressions to illustrate the severity of their pain. An example is the 'Wong Baker Faces Pain Rating Scale'⁵ which uses six different facial expressions from happy (no pain) to agony (the worst pain) to indicate how much pain a person feels (see Table 1). Body diagrams may also be used to help pinpoint where the pain occurs.¹ Simple assessment tools can be useful for assessing pain in people with barriers to communication including children, the elderly, and people from a non-English speaking background (see Table 1).

Pain is multifactorial and is influenced by psychological and environmental factors including a person's previous experiences of pain, and their beliefs, mood and coping ability. The perception and experience of pain is subjective and varies between individuals. There is also some individual variation in response to pain-relieving treatments.

Management of pain

Management of acute pain is directed at symptom relief and treatment of the underlying disorder where possible.^{1,2} Acute pain is expected to be of limited duration and resolve as the injured tissue heals. Non-prescription analgesics may be used 'as needed' for relief of mild to moderate pain and to improve mobility until the pain resolves. Neuropathic pain is less likely to respond to conventional analgesics (e.g. paracetamol, NSAIDs and opioids).⁴

Practice point 1

The 'LINDOCARRF' mnemonic for describing aspects of pain:

- L** – Location(s)
(Where do you feel the pain?)
- I** – Intensity
(Is it mild, moderate, or severe?)
- N** – Nature
(e.g. throbbing, sharp, dull, burning)
- D** – Duration
(How long does the pain last?)
- O** – Occurrence
(When does the pain occur?)
- C** – Concurrence
(Other associated symptoms? e.g. nausea, numbness or weakness)
- A** – Aggravating factors
(What makes the pain worse?)
- R** – Relieving factors
(What makes the pain better?)
- R** – Radiating
(Does the pain travel to other body parts?)
- F** – Frequency
(How often does the pain occur? Is it constant or intermittent?)

Table 1. Simple pain assessment tools⁵

Numeric Rating Scale

0	1	2	3	4	5	6	7	8	9	10
No pain										Unbearable/ worst pain ever)

Wong-Baker FACES Pain Rating Scale



From Hockenberry MJ, Wilson D, Winkelstein ML: *Wong's Essentials of Pediatric Nursing*, ed. 7, St. Louis, 2005, page 1259. Used with permission. Copyright, Mosby.

Practice point 2

Codeine-containing non-prescription pain relievers^{4,6,24}

Codeine is available in combination with paracetamol, aspirin or ibuprofen as *Pharmacist Only* medicine for the temporary relief of moderate to severe acute pain. It is uncertain whether codeine in the amounts contained in these preparations (up to 15 mg per dosage unit) is sufficient to provide additional pain relief compared to paracetamol or NSAIDs alone. It is generally accepted that doses less than 30 mg codeine are subtherapeutic. In addition, approximately 7%–10% of Caucasians and 1%–2% of Asians are 'poor metabolisers' of codeine to morphine (codeine's active metabolite) and obtain negligible analgesic benefit from codeine. These individuals may require alternative analgesic therapy. Codeine has a limited role in chronic pain due to the short duration of action of both codeine and morphine (approximately three hours). Regular use of codeine is commonly associated with drowsiness and constipation. Tolerance, and physical or psychological dependence may also develop with prolonged use. Dependency is more likely in patients with a history of drug or alcohol addiction. Pharmacists should be aware of signs of addictive behavior including repeated requests for codeine-containing analgesics (e.g. because the product has been lost) and attempts to obtain products from multiple sources.

Chronic pain can be caused by a wide variety of conditions and requires clinical assessment and management by a doctor. The psychological factors contributing to the person's emotional and physical experience of pain also need to be considered.

Non-prescription pain relievers

Pain relieving medicines available without a prescription include paracetamol and NSAIDs. These medicines are widely used by consumers, often without the knowledge and advice of a health care professional. Pharmacists can play an important role in advising consumers on the safe and appropriate use of non-prescription analgesics for the management of acute pain, as well as recognising when referral to a doctor or other health care professional is necessary.

Paracetamol and NSAIDs are suitable for short term relief of mild to moderate pain. The most appropriate analgesic for an individual depends on the likely cause of the pain, the presence of risk factors for NSAID-related adverse effects and the relative risk of adverse effects with different NSAIDs. Non-prescription analgesics are frequently used for symptomatic relief of acute pain associated with the following conditions:²

- postoperative pain
- musculoskeletal pain (e.g. shoulder, neck or back pain)
- primary dysmenorrhea
- headache and migraine
- cold and flu symptoms (e.g. sore throat, sinus pain)
- dental pain.

Combination analgesics containing codeine may be recommended for temporary relief of moderate to severe acute pain (see Practice Point 2, and PSA's *Guidance document on the 'Provision of a combination analgesic containing codeine as a Pharmacist Only medicine'* for more information). Chronic pain may also be managed with certain non-prescription medicines under the supervision of a doctor, with regular review to ensure pain is being adequately controlled.

Paracetamol

Paracetamol is an effective analgesic for mild to moderate pain. The incidence of adverse effects at recommended doses is comparable to placebo.² It can be used when NSAIDs are contraindicated or pose an unacceptable risk of side effects.

The recommended dosage in adults and children over 12 years of age is 1–2 tablets (500–1000 mg) every 4–6 hours, up to a maximum of 4 g (8 tablets) per day. If using controlled-release tablets, the dose is 2 tablets (1330 mg) every 6–8 hours,



Paracetamol and NSAIDs are suitable for short term relief of mild to moderate pain.

up to a maximum of six tablets per day. For children over one month of age, dose calculations are based on weight: 15 mg/kg every 4–6 hours up to a maximum of 60 mg/kg per day.⁶ Higher doses up to 90 mg/kg/24 hours may be prescribed for up to 48 hours under medical supervision.⁷ Paracetamol has been shown to have a dose-response effect, with doses of 1000 mg found to be superior to 500 mg in adults.² Paracetamol is also an effective adjunct to opioid analgesia. Opioid requirements are reduced by 20–30% when combined with regular doses of paracetamol.²

Safety precautions

Regular dosing for greater than 24–48 hours is not recommended in children and adolescents without medical review. There is an increased risk of paracetamol toxicity in children and adolescents with fever or poor oral intake, even at doses close to therapeutic doses, when given for more than 48 hours.⁸

The routine use of prophylactic paracetamol at the time of infant vaccination is no longer recommended, as it has been associated with reduced antibody response. Paracetamol (or other analgesics or antipyretics) should only be used if post-vaccine symptoms develop.^{9,10}

In adults, repeated ingestion of supratherapeutic (excessive) doses can cause liver injury. When advising on the safe and correct use of paracetamol, it is important to clarify whether people are taking other products containing paracetamol and the total daily dose should be calculated where necessary. Paracetamol is involved in a number of cases of poisoning requiring hospitalisation

due to inadvertent, repeated supratherapeutic dosing. The risk of liver injury in overdose is increased by chronic alcohol abuse, hepatic impairment (e.g. viral hepatitis, alcoholic hepatitis) concurrent use of some microsomal-inducing drugs (e.g. barbiturates, carbamazepine, rifampicin, isoniazid) and glutathione depletion (e.g. recent prolonged fasting, acute illness with prolonged vomiting or dehydration, anorexia nervosa, bulimia, malnutrition, malignancy, HIV-AIDS).⁸ Therapeutic doses of paracetamol are unlikely to cause hepatotoxicity in patients who ingest moderate to large amounts of alcohol.¹¹ In patients with chronic alcoholism or liver disease, paracetamol may be preferred to NSAIDs but should only be used under medical supervision. Careful dosage instructions are necessary to minimise the potential for accidental overdose.

Regular doses of paracetamol for prolonged periods can increase the anticoagulant effect of warfarin. Increased monitoring and dose adjustment of warfarin may be necessary, particularly for patients who are susceptible to changes in their INR (e.g. the elderly), and those at increased risk or with a history of significant bleeding. Occasional doses of paracetamol (i.e. less than 2 g/day for one day) are generally considered safe in people taking warfarin, with no additional monitoring required. However, regular or higher doses of paracetamol should only be used on medical advice.¹²

Non-steroidal anti-inflammatory drugs (NSAIDs)

Non-prescription NSAIDs are suitable for temporary treatment of mild to moderate

Practice point 3

Some contraindications and precautions with non-prescription NSAIDs that require referral:^{4,6}

- Pregnancy
- Other medical conditions
 - Ischaemic heart disease, hypertension, heart failure
 - History of stroke or myocardial infarction
 - Renal or hepatic impairment
 - History of allergy to aspirin / NSAIDs (i.e. asthma, angioedema, urticaria)
 - Inflammatory bowel disease
- Over 65 years of age
- Planned surgery or dental surgery
- Poor response
- Prolonged use for more than 10 days.

Table 2. Non-prescription NSAIDs⁶

Drug	Recommended doses for non-prescription use
Aspirin	Adults and children over 16 year of age: 300–900 mg every 4–6 hours (maximum 3600 mg per day).
Diclofenac	Adults and children over 14 years of age: 25 mg initially, then 12.5–25 mg every 4–6 hours (maximum of 75 mg per day). Pharmacists may recommend higher doses for short term treatment (up to one week) of acute pain; consult product information for further information.
Ibuprofen	Adults and children over 12 years of age: 200–400 mg every four hours (maximum of 1200 mg per day). Children from 3 months to 12 years of age: the dose is based on the child's weight, 5–10 mg/kg every 6–8 hours (maximum of 3 doses in 24 hours).
Mefenamic acid	Adults and children over 14 years of age: 500 mg three times a day from the onset of period pain for a maximum of seven days.
Naproxen sodium	Adults: 550 mg initially, then 275 mg every 6–8 hours (maximum of 1375 mg per day).

Practice point 4

Potential drug interactions with NSAIDs:⁶

- Angiotensin converting enzyme (ACE) inhibitor or angiotensin II receptor blocker (ARB) + diuretic
- Other antihypertensives
- Anticoagulant / antiplatelet therapy (including low dose aspirin)
- Bisphosphonates
- Corticosteroids
- Lithium
- Selective serotonin reuptake inhibitors (SSRIs) / Selective Noradrenaline Reuptake Inhibitors.

Related Fact Cards

- » Back Pain
- » Colds and Flu
- » First Aid in The Home
- » Gout
- » Headache
- » Migraine
- » Opioids for Pain Relief
- » Osteoarthritis
- » Pain Relievers
- » Period Problems
- » Sprains and Strains

acute pain, especially where there is inflammation associated with tissue injury. NSAIDs may be the first choice for certain painful conditions such as dysmenorrhoea due to their effect on peripheral prostaglandin synthesis. The minimum effective dose should always be used for the shortest duration necessary to control symptoms. NSAIDs given in addition to paracetamol can improve analgesia compared with paracetamol alone.²

Safety precautions

All NSAIDs, including those which can be purchased without a prescription, belong to the same therapeutic class and have similar efficacy. However, at equipotent doses there is some variability in individual response to the effects of different NSAIDs in terms of clinical efficacy and adverse effects. There are some known differences between individual NSAIDs with respect to their anti-inflammatory effects and toxicity. These differences may influence the choice of NSAID used.

Some potential adverse effects and important safety precautions are common to all non-prescription NSAIDs including effects on platelet aggregation and potential to prolong bleeding time. People taking NSAIDs should be advised to seek medical advice before planned surgery. There are important drug interactions with antithrombotic (anticoagulant and antiplatelet) therapies. All NSAIDs have the potential to worsen asthma in people with NSAID-sensitive asthma. NSAIDs trigger asthma symptoms in up to 11% of adults and 2% of children with asthma. Symptoms tend to occur within 1–3 hours of taking a dose, and include runny nose, shortness of breath and wheezing. NSAIDs are not contraindicated if a person with asthma has taken NSAIDs previously without any worsening of their asthma symptoms.¹³ NSAIDs should also be avoided in pregnancy. See Practice Point 3 for a list of common precautions and contraindications to the use of non-prescription NSAIDs.

The gastrointestinal effects of NSAIDs vary from dyspepsia, heartburn, and abdominal discomfort to more serious ulceration with potentially life-threatening bleeding and perforation. Serious gastrointestinal bleeding or perforation is rare and mainly occurs in people with risk factors for NSAID-related gastrointestinal bleeding.

The presence of multiple risk factors greatly increases the risk of gastrointestinal complications.¹⁴ People with any of the following risk factors for gastrointestinal bleeding should be advised to avoid using NSAIDs without medical advice:^{14,15}

- Age ≥ 65 years
- History of peptic ulcer disease
- *Helicobacter pylori* infection
- Multiple NSAIDs
- Prolonged use of NSAIDs at maximal doses
- Anticoagulant or antiplatelet therapy (including low-dose aspirin)
- Oral corticosteroids.

The presence of serious co-morbidity including cardiovascular disease, renal or hepatic impairment, diabetes or hypertension, is an additional risk factor for NSAID-related gastrointestinal complications.¹⁵

Ulcer complications are often asymptomatic. Conversely, dyspepsia and other gastrointestinal symptoms are not always a reliable indicator of mucosal damage.¹⁵ However, people with signs and symptoms suggestive of NSAID-related gastroesophageal irritation require referral. These include:^{14,15}

- Dyspepsia
- Heartburn
- Bloating or cramping
- Nausea and vomiting
- Retrosternal or substernal chest pain
- Difficulty or pain on swallowing
- Vomiting blood or blood in faeces
- Unexplained weight loss.

Comparative information

The relative gastrointestinal and cardiovascular safety of the non-prescription NSAIDs remains uncertain.

Gastrointestinal safety

In healthy individuals with no major risk factors for gastrointestinal bleeding, non-prescription NSAIDs are well tolerated when used short term for up to 10 days at approved doses.^{15–17} Ibuprofen is the most widely used over-the-counter NSAID. In healthy individuals without risk factors, one study found ibuprofen (up to 1.2 g /day) to have similar gastrointestinal tolerability to paracetamol (up to 3 g /day), and better tolerability than aspirin (up to 3 g/day) when used short term for up to



seven consecutive days. Tolerability was evaluated according to the frequency of gastrointestinal adverse events including dyspepsia, nausea, diarrhoea, and abdominal pain.¹⁶ Based on prescription data, ibuprofen (at doses up to 1.2 g/day) has the lowest risk of gastrointestinal complications while diclofenac and naproxen have intermediate risks.^{15,17} However, information on the comparative rates of gastrointestinal adverse events with NSAIDs at approved non-prescription doses is limited.

The gastrointestinal tolerability of ibuprofen and other NSAIDs is dose dependent and the risk of gastrointestinal bleeding increases in a dose-dependent manner.^{15,19,20} Research has found that people often exceed the recommended doses of analgesics and take more in an attempt to obtain quicker or stronger pain relief.¹⁸ It is important that people are warned not to exceed the maximum recommended dose and to limit their use of non-prescription NSAID to no more than 10 days duration without advice from a doctor.

The risk of gastrointestinal adverse effects is known to vary between individual NSAIDs and appears to be influenced by several factors:^{15,16,20,21}

- the dose – the risk of gastrointestinal complications increases at higher doses

- duration of treatment – treatment with NSAIDs for more than 1–3 months can cause damage to the gastrointestinal mucosa, especially in people with risk factors for gastrointestinal effects
- the half-life– NSAIDs with a long half-life (or slow-release formulations) are associated with an increased risk of gastrointestinal complications compared to those with a short half-life
- physiochemical properties of the agent (see Practice Point 6 for more information).

When low-dose aspirin is taken in combination with other NSAIDs, the risk of developing serious gastrointestinal complications increases significantly.^{14,15}

Cardiovascular safety

There is limited information on the cardiovascular safety of NSAIDs at recommended doses for non-prescription use. Most studies have examined the comparative risks with NSAIDs at typical prescription doses.

A 2011 study found the use of prescribed NSAIDs to be associated with an increased risk of death and recurrent myocardial infarction (MI) in patients with a history of MI. The risk of cardiovascular events was independent of treatment duration, increasing almost immediately with diclofenac and COX-2 inhibitors. The contribution of other cardiovascular risk factors (e.g. blood pressure, smoking, obesity) and the concurrent use of non-prescription NSAIDs such as ibuprofen were not examined.²² NSAID use has also been found to increase cardiovascular risk in healthy individuals. Naproxen and low-dose ibuprofen are thought to have a lower cardiovascular risk than other NSAIDs.²³ There appears to be a dose-response relationship and more research is necessary to examine the effects of dose, treatment duration, and patient characteristics on the risk of cardiovascular events.^{22,23} More research is also needed to assess the comparative safety of different non-prescription NSAIDs, which are generally used at lower doses and for short durations.

NSAIDs should be avoided or used with caution in established cardiovascular disease (e.g. previous MI or heart failure). If NSAID therapy is considered necessary, treatment should be limited to the shortest duration possible. Long term use of

Practice point 5

Relative efficacy of non-prescription analgesics^{17,26}

There is limited information available on the equipotent doses of non-prescription pain relievers. The Oxford league table can be used to compare the relative efficacy of different analgesics in acute pain. Analgesic efficacy is expressed as the Number Needed to Treat (NNT), the number of patients who need to receive a drug for one patient to achieve at least 50% relief of moderate to severe acute pain over a 4–6 hour treatment period compared with placebo. The table reflects the dose response to different doses of non-prescription drugs with the most effective drugs having a low NNT of just over 2. For example, ibuprofen 400 mg is approximately equivalent diclofenac 50 mg for relief of moderate to severe pain. When interpreting the information from the table, it is important to note that treatment efficacy varies with different pain symptoms. For example, NSAIDs are more effective for relief of acute pain associated with primary dysmenorrhea compared to paracetamol.

Table. 4 Information from the 'Oxford league table of analgesic efficacy'²⁶

Analgesic	NNT
Diclofenac 50 mg	2.3
Ibuprofen 400 mg	2.4
Ibuprofen 200 mg	2.7
Diclofenac 25 mg	2.8
Naproxen sodium 550 mg	3.0
Paracetamol 1000 mg	3.8

Practice point 6

Gastrointestinal adverse effects of NSAIDs^{14,15}

The gastrointestinal adverse effects of NSAIDs are thought to be mediated by two different mechanisms:

- local injury to the gastrointestinal tract mucosa from direct contact between the drug and mucosa; and
- systemic inhibition of COX enzyme which reduces the cytoprotective effects of prostaglandins.

The relative gastrointestinal safety of individual NSAIDs is thought to be due to differences in inhibition of different isoforms of COX enzyme, particularly COX-1 and COX-2. COX-1 mediates the formation of prostaglandins which are involved in gastrointestinal mucosal protection. Inhibition of these prostaglandins results in decreased synthesis of gastric mucus and bicarbonate, impaired mucosal blood flow, and increased acid secretion. COX-2 catalyses the synthesis of prostaglandins that act as inflammatory mediators in response to tissue injury. COX-2 selective NSAIDs have an improved upper gastrointestinal safety profile.

The potential for NSAIDs to cause gastrointestinal injury is also dependent on their physicochemical properties including their acidic properties (as measured by pKa). These properties determine their topical irritancy in the gastrointestinal mucosa as well as their systemic bioavailability. Gastric injury has been found to correlate significantly with the pKa of the compound: the lower the acidity of the drug, the less the mucosal damage. For example, the selective COX-2 inhibitor celecoxib is non-acidic with a high pKa value. This may account for the low rate of mucosal irritancy, separate from its low inhibitory activity on COX-1 in the stomach. Most non-selective NSAIDs are organic acids with a low pKa (2.8–4.4) which interact with phospholipids causing disruption to the gastric mucosal barrier. Ibuprofen has a pKa of 5.2, while aspirin, indomethacin and naproxen have pKa values ranging from 3.5 to 4.2. This may help to explain the differences in gastrointestinal tolerability and ulcerogenic activity between these NSAIDs.

NSAIDs with low dose aspirin may reduce the cardioprotective benefits of low dose aspirin.¹⁷ This combination should only be used on medical advice.

Other treatment options for management of pain

Other options for management of acute pain include heat packs, topical creams and ointments, rest, and physical therapy.

Topical NSAIDs^{2,6}

Available topical NSAIDs include benzydamine, diclofenac, ibuprofen, ketoprofen and piroxicam. These are available as topical creams/gels for application to the skin. They are normally massaged gently into the affected areas two to four times a day for the short term treatment (1–2 weeks) of musculoskeletal inflammation and injury such as sports injuries, sprains, tendonitis and swelling.

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Complementary and alternative medicine (CAM) therapies

CAM therapies are becoming increasingly popular for self-treatment of common health problems including pain. These therapies may be used on their own or in addition to conventional medicines and they are often used without the knowledge or advice of health care professionals. This increases the risk of side effects and drug interactions. Effective communication between pharmacists and consumers about their self-treatment choices is important to ensure the use of CAM therapies is safe and appropriate. Herbal products and supplements commonly used for self-treatment of pain include arnica, devils claw, evening primrose oil, fish oil/omega-3 oils, garlic, glucosamine +/- chondroitin, MSM, S-adenosylmethionine (SAMe) and willow bark.²⁴

Gastrointestinal Tract 3rd edn [online]. 2009. At: www.gesa.org.au/files/editor_upload/File/DocumentLibrary/Professional/NSAIDS_Clinical.pdf

Assessment questions for the pharmacist

Select one correct answer from each of the following questions.

Answers due 31 May 2012.

Before undertaking this assessment, you need to have read the Facts Behind the Fact Card article in *inPHARMation*, and the associated Fact Cards. This activity has been accredited by PSA as a Group 2 activity. Two CPD credits (Group 2) will be awarded to pharmacists with eight out of 10

questions correct. PSA is authorised by the Australian Pharmacy Council to accredit providers of CPD activities for pharmacists that may be used as supporting evidence of continuing competence.

Submit answers online

To submit your response to these questions online, go to the PSA website www.psa.org.au/selfcare



Accreditation number: CS120003

This activity has been accredited for Group 2 CPD (or 2 CPD credits) suitable for inclusion in an individual pharmacist's CPD plan.

1. Which of the following are risk factors for developing chronic pain?

- a. Poor management of acute pain.
- b. Severe acute pain.
- c. Nerve injury.
- d. All of the above.

2. Choose the CORRECT statement about paracetamol.

- a. Less than 3 g/day for one day is generally considered safe in people taking warfarin.
- b. Paracetamol is contraindicated in people who ingest moderate to large amounts of alcohol.
- c. Paracetamol is generally safe in overdose.
- d. Routine use of paracetamol at the time of infant vaccination is no longer recommended.

3. Choose the INCORRECT statement about NSAIDs.

- a. There is improved pain relief when NSAIDs and paracetamol are taken together.
- b. NSAID are contraindicated in people with a history of asthma.
- c. All non-prescription NSAIDs can affect platelet aggregation and prolong bleeding time.
- d. NSAIDs should be used with caution in people taking corticosteroids.

4. NSAIDs should only be used on medical advice in:

- a. Ischaemic heart disease.
- b. Patients with a history of stroke.
- c. Inflammatory bowel disease.
- d. Planned surgery or dental surgery.
- e. All of the above.

5. Choose the INCORRECT statement about NSAID-related gastrointestinal toxicity.

- a. When low-dose aspirin is taken in combination with NSAIDs the risk of serious gastrointestinal events is decreased.
- b. There is a low risk of gastrointestinal complications when non-prescription NSAIDs are used short term for up to 10 days at approved doses.
- c. The gastrointestinal tolerability of NSAIDs is dose dependent and the risk of gastrointestinal bleeding increases in a dose-dependent manner.
- d. Treatment with NSAIDs for more than 1–3 months can cause damage to the gastrointestinal mucosa, especially in people with risk factors for gastrointestinal effects.

6. Which of the following is a possible sign of NSAID-related gastroesophageal irritation?

- a. Difficulty or pain on swallowing.
- b. Retrosternal or substernal chest pain.
- c. Dyspepsia.
- d. Bloating or cramping.
- e. All of the above.

7. Choose the INCORRECT statement about NSAID-related cardiovascular risk.

- a. NSAIDs should be avoided or used with caution in established cardiovascular disease.
- b. Long term use of NSAIDs in combination with low dose aspirin may improve the cardioprotective benefits of low dose aspirin.
- c. The risk of cardiovascular events associated with NSAID use is thought to be dose dependent.
- d. NSAIDs increase the risk of recurrent myocardial infarction (MI) in patients with a history of MI.

8. Which of the following medicines should be used with caution in combination with NSAIDs?

- a. Venlafaxine.
- b. Low dose aspirin.
- c. Bisphosphonates.
- d. Lithium.
- e. All of the above.

9. Choose the INCORRECT statement about gastrointestinal toxicity caused by NSAIDs.

- a. Local injury to the gastrointestinal tract mucosa contributes to the gastrointestinal adverse effects of NSAIDs.
- b. Systemic inhibition of cyclooxygenase (COX-1) enzyme by NSAIDs reduces the cytoprotective effects of prostaglandins.
- c. NSAIDs with low pKa values (2.8–4.4) have lower rates of gastrointestinal irritancy.
- d. Prostaglandins stimulate mucus and bicarbonate production and decrease acid secretion in the gastrointestinal tract.

10. Which of the following factors influences the risk of gastrointestinal adverse effects with different NSAIDs?

- a. Dose.
- b. Presence of risk factors.
- c. Treatment duration.
- d. Half-life.
- e. All of the above.